

PRINCIPLES OF ECONOMICS

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CHAPTER 2

Thinking Like an Economist

Complete Study Note

8th Edition

This study note covers all major topics from Chapter 2:

**Scientific Method | Economic Models | Circular-Flow Diagram | PPF | Micro vs Macro |
Positive vs Normative**

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1. Introduction

Chapter 2 of Mankiw's **Principles of Economics** introduces the way economists think about the world. Economics has its own language and its own way of looking at problems. Just as a doctor uses medical terms or a lawyer uses legal terms, an economist uses special ideas and tools.

This chapter answers a key question: **What does it mean to think like an economist?**

The chapter is divided into four main parts:

- **Part 1: The Economist as Scientist** - How economists study the economy using scientific methods and models.
- **Part 2: The Economist as Policy Adviser** - How economists give advice using positive and normative analysis.
- **Part 3: Why Economists Disagree** - The two main reasons economists give different advice.
- **Part 4: Let's Get Going** - A brief call to start using economic thinking.

Key Note: After reading this note, you will understand how economists think, how they build and use models, and how they differ from one another.

2. The Economist as Scientist

Economists try to study the economy the same way a scientist studies nature. A physicist studies matter. A biologist studies life. An economist studies how people and societies make decisions about resources.

The key tool is the **scientific method**: observing facts, building a theory, testing the theory with data, and revising the theory if needed.

2.1 The Scientific Method: Observation, Theory, and More Observation

Science works in a cycle. It starts with **observation**. Then a **theory** is built to explain what is observed. Then the theory is tested with more **observation**.

Real-Life Example: Isaac Newton and Gravity

Newton saw an apple fall from a tree. He observed this event and then built a theory about gravity. Later, scientists tested his theory and found it worked well in many situations. Economics uses the same approach.

Real-Life Example: Inflation Theory

An economist observes that prices are rising quickly in a country. The economist builds a theory: 'High inflation happens when the government prints too much money.' The economist then collects data from many countries. If countries with more money growth also have higher inflation (as is actually true), the theory is supported.

A Challenge for Economics: In science, experiments can be done in a laboratory. A physicist can drop objects to test gravity. But economists cannot experiment on a whole economy. They cannot change the money supply of a country just to collect data.

Instead, economists use **natural experiments** - events that happen in history that act like experiments.

Real-Life Example: Natural Experiment: Middle East Oil Shocks

When war disrupts oil supplies from the Middle East, oil prices rise sharply around the world. Economists cannot cause this event, but they can study it to understand how oil prices affect economies.

2.2 The Role of Assumptions

Definition: Assumption

A simplification that economists make to study a complex problem more easily. An assumption removes details that are not important for the question being asked.

Assumptions help simplify the real world so that we can understand the most important parts of a problem. Without assumptions, the world would be too complex to study.

Real-Life Example: Physicist's Assumption

A physicist calculating how fast a marble falls from a building assumes there is no air friction. In reality, there is air. But the friction on a marble is so small that ignoring it does not change the answer much. The same assumption does NOT work for a beach ball, because air friction matters a lot for a large object.

Why Economists Use Different Assumptions for Different Questions:

Suppose the government changes the amount of money in circulation. How do prices respond? This depends on the time period:

- **Short run:** Economists assume that prices do not change much. Some even assume prices are completely fixed.
- **Long run:** Economists assume that prices are fully flexible and can adjust to any level.

Key Note: The art of economics is knowing which assumptions to use for each question.

2.3 Economic Models

Definition: Economic Model

A simplified representation of the economy. It uses diagrams or equations to show the most important features of an economic situation, leaving out unnecessary details.

Just like a biology teacher uses a plastic model of the human body to explain how organs work, an economist uses diagrams and equations to explain how the economy works.

A model of the human body shows the heart, liver, and kidneys, but it does not show every muscle or blood vessel. This simplification makes learning easier. Economic models work the same way: they show the most important parts of the economy without every detail.

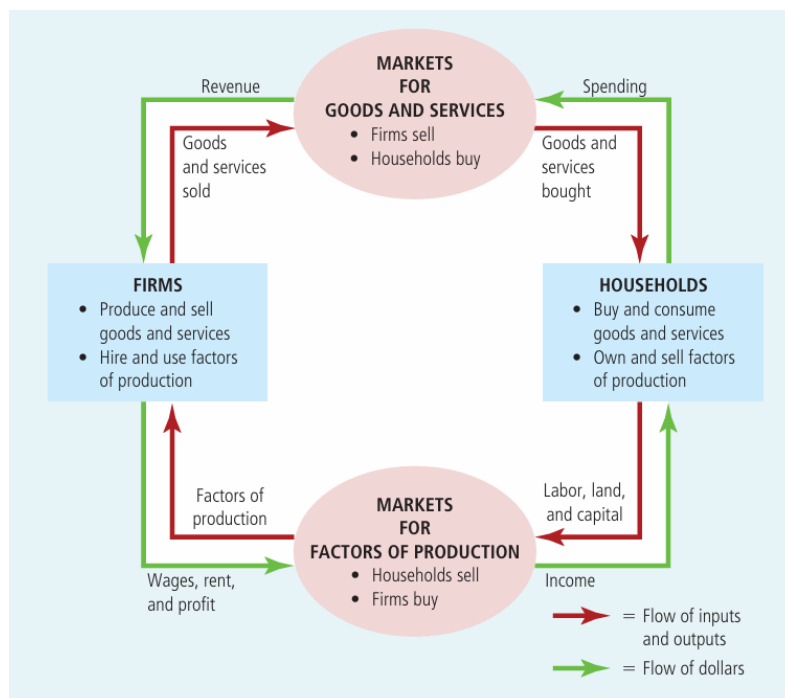
Key Note: All models - in physics, biology, and economics - simplify reality to improve our understanding of it.

2.4 Model 1: The Circular-Flow Diagram

Definition: Circular-Flow Diagram

A visual model of the economy that shows how money (dollars) flows between households and firms through two types of markets: the market for goods and services, and the market for factors of production.

The economy involves millions of people doing many things. The circular-flow diagram simplifies this into two main groups and two main markets.



The Two Groups

Group	What They Do	What They Own/Produce
FIRMS	Produce goods and services. Hire workers and use land and capital.	They own the production process. They produce outputs.
HOUSEHOLDS	Buy goods and services from firms. Provide labor, land, and capital to firms.	They own the factors of production (labor, land, capital).

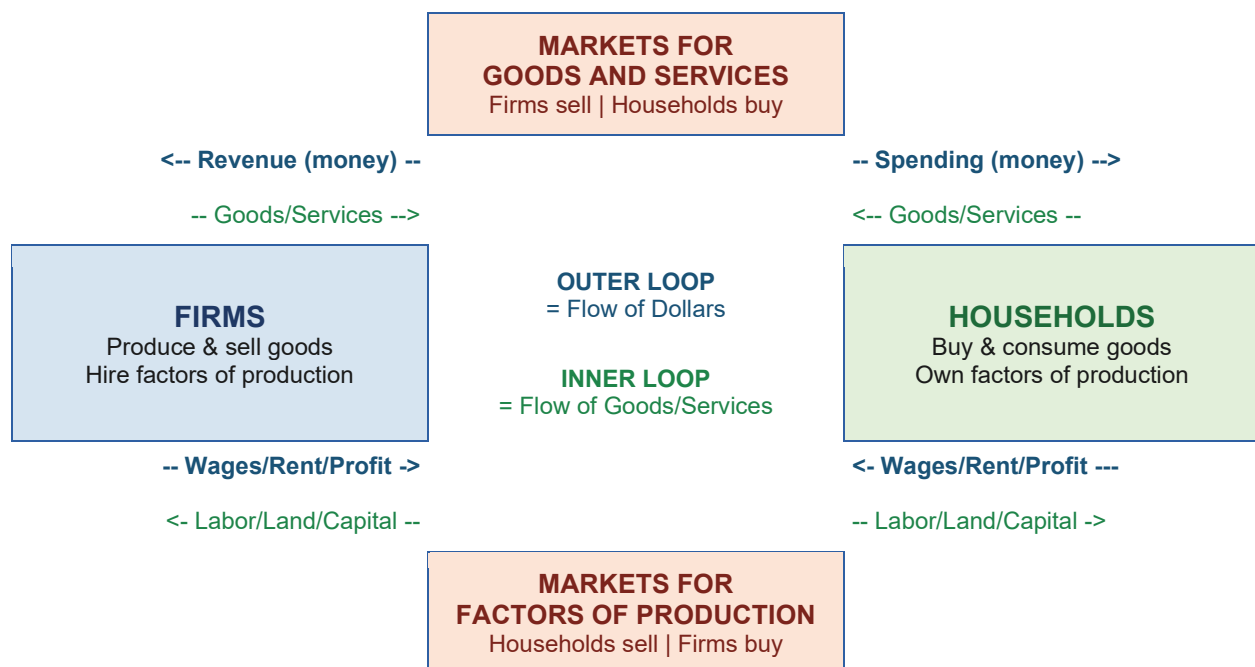
The Two Markets

Market	Who Sells?	Who Buys?	What is Exchanged?
Market for Goods & Services	Firms	Households	Products: food, clothes, electronics, etc.
Market for Factors of Production	Households	Firms	Labor, land, and capital

The Two Loops

- **Inner Loop (Real Goods and Services):** Shows the physical flow. Households give labor, land, and capital to firms. Firms give goods and services back to households.
- **Outer Loop (Money / Dollars):** Shows the money flow. Households pay money to firms for goods. Firms pay wages, rent, and profit to households.

Circular-Flow Diagram



Walking Through the Circular Flow: A Dollar Bill Example

Imagine you have a dollar bill in your wallet. You go to a coffee shop and buy a coffee. The dollar enters the shop's cash register. The shop uses the dollar to pay its workers or pay rent.

The worker or landlord now holds the dollar. They then spend it on other goods. The dollar keeps moving in a circle.

Key Note: This model is simple. In reality, governments and foreign trade also play a role. But this basic model helps us understand the core structure of any economy.

2.5 Model 2: The Production Possibilities Frontier (PPF)

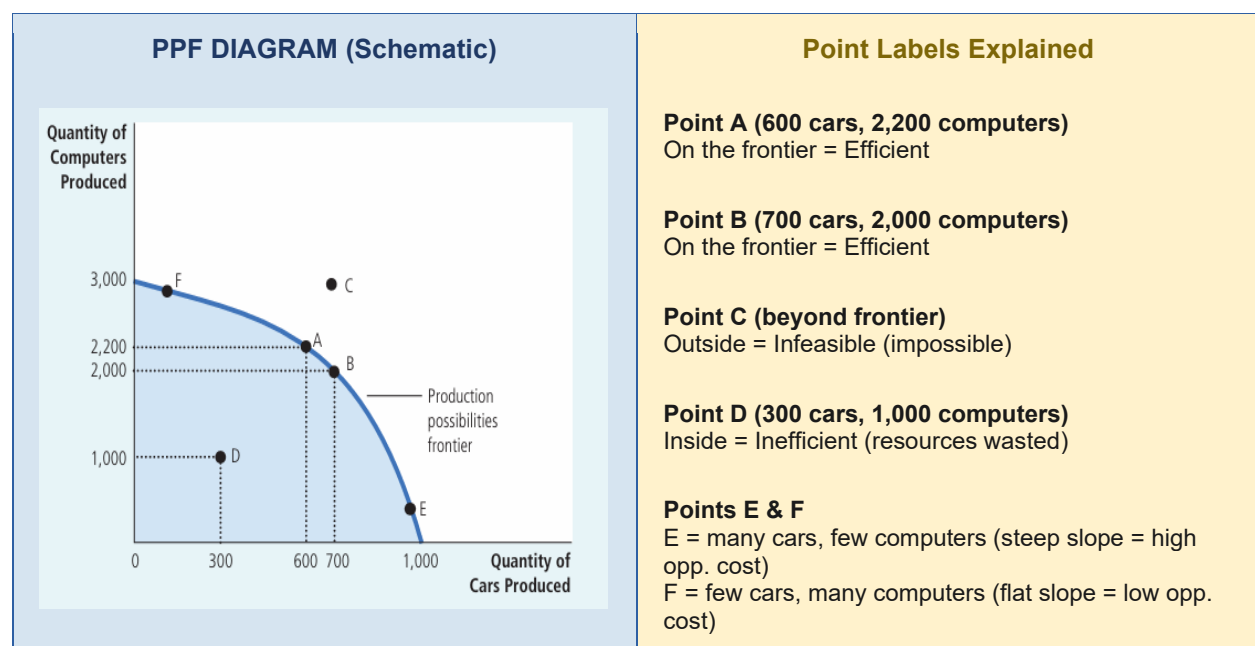
Definition: Production Possibilities Frontier (PPF)

A graph that shows all the possible combinations of two goods that an economy can produce, given its available resources and technology.

To keep things simple, imagine an economy that makes only two goods: **cars** and **computers**. All resources (workers, machines, land) are used to make one or both of these goods.

The PPF shows the maximum output combinations. If all resources are used for cars, the economy can make 1,000 cars and 0 computers. If all resources are used for computers, the economy can make 3,000 computers and 0 cars. Between these extremes are many other combinations.

PPF Diagram



Key Concepts from the PPF

Definition: Efficiency

A situation where the economy is getting all it can from its available resources. On the PPF, every point on the curve itself is efficient.

Definition: Inefficiency

A situation where the economy is producing less than it could with available resources. Points inside the PPF curve are inefficient.

Definition: Infeasibility

A situation where the desired production level is not possible with current resources and technology. Points outside the PPF are infeasible.

Definition: Opportunity Cost

The cost of getting something measured by what you give up. On the PPF, producing more of one good means giving up some of the other good.

Trade-offs and Opportunity Cost on the PPF

Moving from **Point A** (600 cars, 2,200 computers) to **Point B** (700 cars, 2,000 computers) means:

- Society produces 100 MORE cars.
- Society produces 200 FEWER computers.
- The opportunity cost of 100 cars = 200 computers.
- The opportunity cost of 1 car = 2 computers.

Why the PPF Curves Outward (Bowed Shape)

The PPF is not a straight line. It bows outward. This happens because not all resources are equally good at producing both goods.

- **At Point F** (many computers, few cars): Resources best suited for cars are being used in the computer industry. Moving a few of them into car production is easy and cheap. The opportunity cost of a car is LOW. The curve is relatively flat here.
- **At Point E** (many cars, few computers): The best car workers are already in the car industry. To produce one more car, you must move skilled computer technicians away

from computers. This causes a big loss of computers. The opportunity cost of a car is HIGH. The curve is steep here.

Key Note: The bowed shape of the PPF shows increasing opportunity cost: as you produce more of one good, it becomes more and more costly to produce additional units of it.

Economic Growth and the PPF

Over time, the PPF can shift **outward**. This means the economy can now produce more of both goods. This happens because of:

- Technological advances (e.g., a better computer-making machine)
- More resources (e.g., more workers or more capital)
- Better education and training

Real-Life Example: Technology Shift

If a new technology is invented that helps workers make computers faster, the computer industry can produce more with the same amount of resources. The PPF shifts outward. Society can now move from Point A (600 cars, 2,200 computers) to Point G (650 cars, 2,300 computers), getting more of BOTH goods.

Summary: What the PPF Teaches Us

Concept	What the PPF Shows
Scarcity	Resources are limited; not every combination is possible.
Efficiency	Points on the curve use all available resources fully.
Trade-offs	More of one good means less of the other.
Opportunity Cost	The slope of the curve shows what is given up.
Economic Growth	New technology or resources shift the curve outward.

2.6 Microeconomics and Macroeconomics

Economics is divided into two major fields, just as biology is divided into molecular biology, cellular biology, and evolutionary biology. The two fields of economics are **microeconomics** and **macroeconomics**.

Definition: Microeconomics

The study of how individual households and firms make decisions and how they interact in specific markets.

Definition: Macroeconomics

The study of economy-wide phenomena, including inflation, unemployment, and economic growth.

Feature	MICROECONOMICS	MACROECONOMICS
Focus	Individual people, firms, or markets	The whole economy
Level	Small scale (micro = small)	Large scale (macro = large)
Questions Asked	How do rent controls affect housing? How does competition affect prices?	What causes inflation? What causes unemployment? How can an economy grow?
Examples	Effects of minimum wage on jobs. Why do doctors earn more than teachers?	Government borrowing, national growth rates, trade deficits

Even though microeconomics and macroeconomics study different levels, they are **connected**. Large economic changes (macroeconomics) come from millions of small individual decisions (microeconomics). For example, if the government cuts income taxes, households (micro) decide to spend more. This increases total spending in the economy (macro).

3. The Economist as Policy Adviser

Economists play two different roles. When they explain how the economy works, they act as **scientists**. When they recommend what the government should do, they act as **policy advisers**.

3.1 Positive versus Normative Analysis

The most important distinction in policy analysis is between **positive statements** and **normative statements**.

Definition: Positive Statement

A statement that describes how the world IS. It is a factual claim that can be tested with data and evidence. It does not involve value judgments.

Definition: Normative Statement

A statement that describes how the world SHOULD BE. It involves value judgments and opinions about what is good or bad. It cannot be proven true or false only with data.

Comparison: Positive vs. Normative Statements

Feature	POSITIVE STATEMENT	NORMATIVE STATEMENT
Nature	Descriptive: Describes facts	Prescriptive: Suggests what should be done
Can Data Prove It?	Yes. It can be confirmed or disproven by evidence.	No. It depends on values and ethics, not just facts.
Role	Economist as scientist	Economist as policy adviser
Example	"Minimum-wage laws cause unemployment."	"The government should raise the minimum wage."
Example 2	"Higher taxes reduce consumption."	"Taxes on the rich should be increased."

Important note: Positive and normative statements are often connected. If Portia believes that minimum wage **causes unemployment** (positive), she may conclude that the government **should not raise** the minimum wage (normative). But the normative conclusion also depends on her **values** - how much she cares about employment versus helping low-wage workers.

Real-Life Example: Minimum Wage Debate

Portia says: 'Minimum-wage laws cause unemployment.' This is a POSITIVE statement. It claims a fact about the world. We can test it by studying data on unemployment when minimum wages change. Noah says: 'The government should raise the minimum wage.' This is a NORMATIVE statement. It tells us what Noah thinks should be done. It reflects his values about fairness for workers.

3.2 Economists in Washington

Economists work in government to give advice on policy. Some important positions include:

- **Council of Economic Advisers:** A group of three economists who advise the U.S. President on economic policy. They also write the annual Economic Report of the President.
- **Office of Management and Budget:** Economists here help plan government spending and regulations.
- **Department of the Treasury:** Economists here help design tax policy.
- **Federal Reserve:** Employs hundreds of economists to analyze economic conditions and advise on monetary policy.
- **Congressional Budget Office:** Provides independent economic analysis to the U.S. Congress.

3.3 Why Economists' Advice Is Not Always Followed

Even when economists give clear advice, policymakers do not always follow it. There are several reasons:

- Political pressure from different groups with competing interests.
- Public opinion may not support the economists' recommendation.
- Policy decisions involve communication, public relations, and political strategy, not just economics.
- Elected leaders must balance many goals, not just economic efficiency.

Key Note: Economic advice is only one input in the complex process of making policy. Political realities often limit what economists can recommend or implement.

4. Why Economists Disagree

It is common to hear that economists always disagree. George Bernard Shaw famously said: **"If all economists were laid end to end, they would not reach a conclusion."** This section explains why economists sometimes give different advice, and also shows that economists agree far more than people think.

4.1 Two Reasons for Disagreement

Reason 1: Differences in Scientific Judgments

Economists sometimes disagree because they have different **theories about how the world works**, or different estimates of important numbers.

Real-Life Example: Income Tax vs. Consumption Tax

Some economists believe switching from income tax to consumption tax would greatly increase household savings, leading to more investment and growth. Other economists believe savings would not change much with this tax switch. Both groups are making different positive claims about how people respond to taxes. Because of these different views, they give different policy advice.

Reason 2: Differences in Values

Sometimes economists disagree not because of facts, but because of **different values**. They may agree on the facts but disagree on what is fair or desirable.

Real-Life Example: Tax Fairness Example

Peter earns \$150,000 and pays \$15,000 in tax (10%). Paula earns \$30,000 and pays \$6,000 in tax (20%). Is this fair? Some economists value equality of income and say Paula pays too much. Others value equal percentage contribution and think the system is fine. No amount of data can answer this - it depends on values.

4.2 Areas of Agreement Among Economists

Despite disagreements, economists agree on many important topics. Surveys of professional economists show wide agreement on the following propositions:

Proposition	% of Economists Who Agree
Rent control reduces housing quality and availability	93%
Tariffs and import quotas reduce general economic welfare	93%
Flexible exchange rates are an effective monetary arrangement	90%
The U.S. should not restrict outsourcing to foreign countries	90%
Fiscal policy has a significant effect on a less-than-fully employed economy	90%
A large federal budget deficit harms the economy	83%
Inflation is caused primarily by too much growth in the money supply	83%
A minimum wage increases unemployment among young and unskilled workers	79%

Key Note: Most economic disagreements that appear in the news are about complex topics where facts and values are mixed together. On many core economic questions, economists agree strongly.

5. Key Definitions Glossary

This section lists all important terms from Chapter 2 with their definitions.

Term	Definition
Scientific Method	A process of building theories about the world, collecting data, and testing the theories using observations.
Natural Experiment	A real-world event (such as a war or natural disaster) that allows economists to study cause-and-effect relationships without running a controlled lab experiment.
Assumption	A simplification used to make a complex problem easier to study. Assumptions remove unimportant details.
Economic Model	A simplified representation of the economy using diagrams or equations to show what is most important.
Circular-Flow Diagram	A visual model showing how money flows between households and firms through goods/services markets and factor markets.
Factors of Production	The inputs used to produce goods and services: labor, land, and capital.
Production Possibilities Frontier (PPF)	A graph showing all combinations of two goods an economy can produce with its available resources and technology.
Efficiency	A state where the economy gets the maximum possible output from its available resources. Points on the PPF are efficient.
Opportunity Cost	The value of what you give up to get something else. On the PPF, it equals the slope of the curve.
Microeconomics	The branch of economics studying individual households, firms, and specific markets.
Macroeconomics	The branch of economics studying economy-wide phenomena: inflation, unemployment, and economic growth.
Positive Statement	A descriptive claim about how the world IS. Can be tested with data.
Normative Statement	A prescriptive claim about how the world SHOULD BE. Involves value judgments.

6. Exam Revision Summary

Use this section for quick review before an exam. It covers the most important points from Chapter 2.

6.1 Quick Review: Key Concepts

Topic	Key Point to Remember
Economist as Scientist	Economists use the scientific method: observe, theorize, test. They use natural experiments because lab experiments are not possible.
Role of Assumptions	Assumptions simplify complex problems. Good economists choose the right assumptions for the right question.
Economic Models	Models are simplified tools (diagrams/equations) to understand the economy. They omit unnecessary details.
Circular-Flow Diagram	Shows money flow between firms and households through two markets: goods/services and factors of production.
PPF	Shows trade-offs, efficiency (on the curve), inefficiency (inside), and infeasibility (outside the curve).
Opportunity Cost	Shown by the slope of the PPF. Increases as you produce more of one good (bowed outward curve).
Economic Growth	Better technology or more resources shifts the PPF outward, allowing more of both goods.
Micro vs. Macro	Micro = individual decisions; Macro = economy-wide patterns. Both are connected.
Positive vs. Normative	Positive = facts (testable); Normative = values (not testable). Key to understanding economists' roles.
Why Economists Disagree	Different positive theories OR different values. But economists agree on many core questions.

6.2 Key Formulas and Relationships

Opportunity Cost of Good X = Slope of the PPF at that point

Movement along the PPF: More of Good A --> Less of Good B (trade-off)

PPF shifts outward when: Technology improves OR Resources increase

PPF shifts inward when: Disaster, war, loss of resources

Outer loop of circular flow: Dollars (money). Households pay firms; firms pay households.

Inner loop of circular flow: Real goods and factors. Firms give goods; households give labor/land/capital.

6.3 Common Exam Questions and Answers

Exam Question	Short Answer
What is the scientific method in economics?	Observe economic events, develop theories, test theories using data. Use natural experiments instead of lab experiments.
Why do economists use assumptions?	To simplify complex problems so the most important relationships can be studied clearly.
What are the two loops in the circular-flow diagram?	Inner loop: flow of goods and factors (real). Outer loop: flow of money (dollars).
What is a point inside the PPF?	An inefficient point. Resources are not being fully used (e.g., due to unemployment).
What is a point outside the PPF?	An infeasible point. The economy does not have enough resources to reach it with current technology.
What does the slope of the PPF represent?	The opportunity cost of producing one more unit of the good on the horizontal axis.
What is the difference between micro and macro?	Micro studies individual decisions; macro studies economy-wide trends like inflation and unemployment.
What is a positive statement? Give an example.	A factual claim testable by data. Example: 'Higher minimum wages increase unemployment.'
What is a normative statement? Give an example.	A value-based claim about what should happen. Example: 'The government should help the poor.'
Why do economists sometimes disagree?	Either they have different positive theories (scientific disagreement) or different values (normative disagreement).

6.4 Important Comparisons to Remember

Scientist vs. Policy Adviser

- Scientist: Describes the world as it IS. Uses positive analysis.
- Policy Adviser: Recommends how the world SHOULD BE. Uses normative analysis.

Positive vs. Normative

- Positive: Testable with evidence. Example: 'Price controls cause shortages.'
- Normative: Based on values. Example: 'We should protect low-income renters.'

Micro vs. Macro

- Micro: Why does bread cost \$2? Why do nurses earn more than bus drivers?
- Macro: Why is inflation 5% this year? Why is unemployment rising?

Efficient vs. Inefficient vs. Infeasible (on PPF)

- On the curve = Efficient (fully using all resources)
- Inside the curve = Inefficient (wasting resources)
- Outside the curve = Infeasible (impossible with current resources)

End of Chapter 2 Study Note

Thinking Like an Economist - Mankiw, Principles of Economics